

Intermediate Probability and Statistics

A Calculus-Based Course with Theory, Examples, and Intuition

Theodosios Dimitrasopoulos*

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*Email: theodosios.novak@gmail.com

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1 How to Use This Course

This document assumes you are comfortable with the basics of probability and statistics: random variables, common distributions (Binomial, Normal, Exponential), expectation, variance, the law of large numbers, simple confidence intervals, hypothesis tests, and simple linear regression.¹

The goal is to deepen your understanding of *why* standard statistical procedures work by exposing the underlying probabilistic and asymptotic theory, while keeping the exposition example-driven and intuition-focused. You will see more formal definitions and theorems, but we will always accompany them with concrete calculations, visual pictures in words, and connections to practical contexts (e.g., finance, simulation, regression modeling).

A reasonable path through the material is:

1. Sections 2–5: probability foundations and asymptotics.
2. Sections 9–12: sampling distributions, estimation, and hypothesis tests.
3. Sections 13–16: Bayesian methods, linear models, and resampling.

You can also treat individual sections as self-contained mini-lectures.

¹The level is comparable to the transition from introductory probability to a first mathematical statistics course or an “intermediate statistics” module.[web:30][web:36][web:41]

2 Probability Foundations at a Deeper Level

2.1 Sigma-Algebras and Probability Measures

We briefly revisit the measure-theoretic foundation of probability, but focus on intuition and practical usage rather than full rigor.

Definition 2.1 (Sigma-algebra). *Let Ω be a non-empty set (the sample space). A collection $\mathcal{F}$ of subsets of Ω is called a **sigma-algebra** if:*

1. $\Omega \in \mathcal{F}$.
2. If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$.
3. If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

Intuitively, $\mathcal{F}$ is the collection of all events we are able (or willing) to assign probabilities to. For finite or countable Ω , we often take $\mathcal{F}$ to be the collection of all subsets of Ω , but for continuous spaces we usually restrict to a suitable sigma-algebra like the Borel sets of $\mathbb{R}$.

Definition 2.2 (Probability measure). *A function $P : \mathcal{F} \rightarrow [0, 1]$ is a **probability measure** if:*

1. $P(\Omega) = 1$.
2. For disjoint events $A_1, A_2, \dots \in \mathcal{F}$,

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n).$$

This measure-theoretic view unifies discrete and continuous probabilities: point masses and density integrals are both special cases of assigning “sizes” to sets in a consistent way.

2.2 Distribution Functions

For a real-valued random variable X , the **cumulative distribution function** (CDF) is

$$F_X(x) = P(X \leq x), \quad x \in \mathbb{R}.$$

Key properties:

- F_X is non-decreasing.
- $\lim_{x \rightarrow -\infty} F_X(x) = 0$, $\lim_{x \rightarrow +\infty} F_X(x) = 1$.
- F_X is right-continuous.

For discrete X , F_X is a step function; for continuous X with pdf f_X , we have

$$F_X(x) = \int_{-\infty}^x f_X(t) dt.$$

Example 2.1 (Mixture distribution). *Suppose a strategy's daily return R is*

$$R = \begin{cases} 0 & \text{with probability 0.9,} \\ Z & \text{with probability 0.1,} \end{cases}$$

where $Z \sim N(0.02, 0.01^2)$ (2% mean, 1% sd). Then R is neither purely discrete nor purely continuous. Its CDF is a mixture:

$$F_R(x) = 0.9 \mathbf{1}_{\{x \geq 0\}} + 0.1 F_Z(x),$$

where F_Z is the normal CDF of Z . This example illustrates that CDFs are the most general description of a distribution.

2.3 Transformations of Random Variables

Let X be a continuous random variable with pdf f_X and $Y = g(X)$ where g is a differentiable monotone function. Then the pdf of Y is

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

Example 2.2 (Log-transform of a positive variable). *Suppose a positive random variable X has pdf*

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

(Exponential(λ)). Consider $Y = \log X$. Then $X = e^Y$, so $g^{-1}(y) = e^y$ and $\frac{d}{dy} g^{-1}(y) = e^y$. Thus

$$f_Y(y) = f_X(e^y) e^y = \lambda e^{-\lambda e^y} e^y, \quad y \in \mathbb{R}.$$

So a log-transform turns an exponential into a distribution with a heavy right tail in y -space; this kind of transformation is useful in log-return modeling and variance-stabilizing transforms.

3 Modes of Convergence

To understand asymptotic results like the law of large numbers and the central limit theorem, we need precise notions of convergence for sequences of random variables.

3.1 Convergence in Probability

Definition 3.1 (Convergence in probability). *A sequence X_n converges to X in probability, written $X_n \xrightarrow{P} X$, if for every $\varepsilon > 0$,*

$$\lim_{n \rightarrow \infty} P(|X_n - X| > \varepsilon) = 0.$$

Intuitively, for large n , X_n is very likely to be close to X . The law of large numbers is an example: the sample mean $\bar{X}_n$ converges in probability to the true mean μ .

Example 3.1 (Sample mean of i.i.d. variables). *Let $X_1, X_2, \dots$ be i.i.d. with $E[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. Define $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. By Chebyshev's inequality,*

$$P(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2/n}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2}.$$

As $n \rightarrow \infty$, the right-hand side goes to 0, so $\bar{X}_n \xrightarrow{P} \mu$. This is a direct, quantitative illustration of convergence in probability.

3.2 Almost Sure Convergence

Definition 3.2 (Almost sure convergence). *We say X_n converges to X almost surely (a.s.), written $X_n \xrightarrow{a.s.} X$, if*

$$P(\{\omega : X_n(\omega) \rightarrow X(\omega) \text{ as } n \rightarrow \infty\}) = 1.$$

This is a stronger notion: for almost every outcome, the sequence of numbers $X_n(\omega)$ converges in the usual deterministic sense.

Remark 3.1. *Almost sure convergence implies convergence in probability, but not conversely. Roughly, a.s. convergence is pointwise (except on a null set), while convergence in probability is about the probabilities of deviations shrinking.*

3.3 Convergence in Distribution

Definition 3.3 (Convergence in distribution). *X_n converges in distribution to X , written $X_n \Rightarrow X$, if*

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$$

for all x where F_X is continuous.

This mode of convergence is purely about the distributions, not about any underlying probability space. The central limit theorem is a convergence-in-distribution statement.

Example 3.2 (Sample mean CLT preview). *Let $X_1, \dots, X_n$ be i.i.d. with mean μ and variance σ^2 . Define*

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}.$$

The central limit theorem says $Z_n \Rightarrow Z$ where $Z \sim N(0, 1)$. This does not mean Z_n eventually equals a normal random variable, but that its distribution becomes closer and closer to the standard normal distribution.

4 Law of Large Numbers Revisited

4.1 Weak Law of Large Numbers (WLLN)

Theorem 4.1 (Weak Law of Large Numbers). *Let $X_1, X_2, \dots$ be i.i.d. with $E[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. Then*

$$\bar{X}_n \xrightarrow{P} \mu.$$

The proof via Chebyshev's inequality was already sketched. The key point is that as n increases, the sample mean's variance shrinks like $1/n$, so it's increasingly concentrated near μ .

4.2 Strong Law of Large Numbers (SLLN)

Theorem 4.2 (Strong Law of Large Numbers (informal)). *Under mild conditions (for example, i.i.d. with finite expectation $E[|X_i|] < \infty$), we have*

$$\bar{X}_n \xrightarrow{a.s.} \mu.$$

This is stronger than the WLLN: for almost every realization, the time average converges to the expected value. Intuitively, if you keep rolling a fair die forever, the long-run average outcome for a single sequence of rolls will be 3.5.

Example 4.1 (Simulated trading P&L). *Suppose you simulate a trading strategy's daily profit and loss (P&L) $X_1, X_2, \dots$ as i.i.d. with $E[X_i] = \mu$. The strong law says that if you look at a single long simulation path, the average P&L*

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

will almost surely approach μ . This justifies thinking of the expected daily P&L as the long-run average performance in a stable market regime.

5 Central Limit Theorem and Delta Method

5.1 Classical Central Limit Theorem

Theorem 5.1 (Central Limit Theorem (Lindeberg–Levy version)). *Let $X_1, \dots, X_n$ be i.i.d. with $E[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. Define*

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}.$$

Then

$$Z_n \Rightarrow N(0, 1).$$

In practice, this means that for large n , the distribution of $\bar{X}_n$ is approximately normal with mean μ and variance σ^2/n . This underlies normal approximations for sample means, standard error formulas, and many test statistics.

Example 5.1 (Average daily return). *Let $R_1, \dots, R_n$ be i.i.d. daily returns with mean μ and variance σ^2 . Then for large n ,*

$$\bar{R}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right).$$

If you estimate μ by $\bar{R}_n$, you can attach an approximate 95% confidence interval:

$$\bar{R}_n \pm 1.96 \frac{\sigma}{\sqrt{n}}.$$

In practice σ is unknown and replaced by the sample standard deviation, leading to a t -approximation.

5.2 Delta Method

The delta method extends the CLT to smooth functions of asymptotically normal estimators.

Theorem 5.2 (Delta Method (first-order)). *Suppose $\sqrt{n}(T_n - \theta) \Rightarrow N(0, \tau^2)$, and g is differentiable at θ with derivative $g'(\theta) \neq 0$. Then*

$$\sqrt{n}(g(T_n) - g(\theta)) \Rightarrow N(0, (g'(\theta))^2 \tau^2).$$

Heuristically, for T_n close to θ , we approximate

$$g(T_n) \approx g(\theta) + g'(\theta)(T_n - \theta).$$

So the variance of $g(T_n)$ is approximately $(g'(\theta))^2$ times the variance of T_n .

Example 5.2 (Variance-stabilizing transform for Poisson mean). *Let $X_1, \dots, X_n$ be i.i.d. $\text{Poisson}(\lambda)$ with $\lambda > 0$. The sample mean $\bar{X}_n$ satisfies*

$$\sqrt{n}(\bar{X}_n - \lambda) \Rightarrow N(0, \lambda),$$

so $\tau^2 = \lambda$. Define $T_n = \bar{X}_n$ and consider $g(\theta) = 2\sqrt{\theta}$, so $g'(\theta) = 1/\sqrt{\theta}$. Then

$$\sqrt{n}(2\sqrt{T_n} - 2\sqrt{\lambda}) \Rightarrow N\left(0, \left(\frac{1}{\sqrt{\lambda}}\right)^2 \lambda\right) = N(0, 1).$$

So $2\sqrt{\bar{X}_n}$ is approximately normal with mean $2\sqrt{\lambda}$ and variance $1/n$. This suggests that $\sqrt{X}$ -type transforms can stabilize variance in Poisson-like count data.

6 Moment Generating and Characteristic Functions

6.1 Moment Generating Functions

Definition 6.1 (Moment generating function). *The **moment generating function** (mgf) of a random variable X is*

$$M_X(t) = E[e^{tX}],$$

for all t where this expectation is finite.

If M_X exists in an open interval around 0, it uniquely determines the distribution of X . Moments are obtained by differentiation:

$$M_X^{(k)}(0) = E[X^k].$$

Example 6.1 (Normal mgf). *If $X \sim N(\mu, \sigma^2)$, then*

$$M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right).$$

We see directly that $M_X'(0) = \mu$ and $M_X''(0) = \mu^2 + \sigma^2$, so $\text{Var}(X) = M_X''(0) - (M_X'(0))^2 = \sigma^2$ as expected.

6.2 Characteristic Functions

The **characteristic function** of X is

$$\varphi_X(t) = E[e^{itX}], \quad t \in \mathbb{R}.$$

Characteristic functions always exist (they are bounded by 1) and also uniquely determine the distribution of X .

One advantage of characteristic functions is that they linearize sums: if X and Y are independent, then

$$\varphi_{X+Y}(t) = \varphi_X(t) \varphi_Y(t).$$

Example 6.2 (Sum of independent normals). *Let $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ be independent. Then*

$$\varphi_X(t) = \exp\left(i\mu_1 t - \frac{1}{2}\sigma_1^2 t^2\right), \quad \varphi_Y(t) = \exp\left(i\mu_2 t - \frac{1}{2}\sigma_2^2 t^2\right).$$

Thus

$$\varphi_{X+Y}(t) = \varphi_X(t) \varphi_Y(t) = \exp\left(i(\mu_1 + \mu_2)t - \frac{1}{2}(\sigma_1^2 + \sigma_2^2)t^2\right),$$

which is the characteristic function of $N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. So the sum of independent normals is normal, with mean and variance added.

7 Multivariate Distributions and the Multivariate Normal

7.1 Joint, Marginal, and Conditional Distributions

For a random vector $\mathbf{X} = (X_1, \dots, X_d)^\top$, the joint distribution can be described via a joint pdf $f_{\mathbf{X}}(\mathbf{x})$ (continuous case) or joint pmf $p_{\mathbf{X}}(\mathbf{x})$ (discrete).

The **marginal** distribution of X_1 is obtained by integrating/summing over the other coordinates:

$$f_{X_1}(x_1) = \int_{\mathbb{R}^{d-1}} f_{\mathbf{X}}(x_1, x_2, \dots, x_d) dx_2 \cdots dx_d.$$

The **conditional** distribution of X_1 given $X_2 = x_2, \dots, X_d = x_d$ has density

$$f_{X_1|X_2, \dots, X_d}(x_1 | x_2, \dots, x_d) = \frac{f_{\mathbf{X}}(x_1, \dots, x_d)}{f_{X_2, \dots, X_d}(x_2, \dots, x_d)},$$

whenever the denominator is positive.

7.2 Covariance Matrix

For $\mathbf{X} = (X_1, \dots, X_d)^\top$ with mean vector $\boldsymbol{\mu} = E[\mathbf{X}]$, the **covariance matrix** is

$$\Sigma = \text{Cov}(\mathbf{X}) = E[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top].$$

The (i, j) entry is $\text{Cov}(X_i, X_j)$. This matrix is symmetric and positive semi-definite.

Example 7.1 (Portfolio variance). Let $\mathbf{R} = (R_1, \dots, R_d)^\top$ be returns of d assets, with covariance matrix Σ , and let $\mathbf{w}$ be a vector of portfolio weights (summing to 1, say). Then the portfolio return is $R_p = \mathbf{w}^\top \mathbf{R}$, and

$$\text{Var}(R_p) = \mathbf{w}^\top \Sigma \mathbf{w}.$$

This formula highlights how both individual variances and cross-asset covariances contribute to risk.

7.3 Multivariate Normal Distribution

Definition 7.1 (Multivariate normal). A random vector $\mathbf{X} \in \mathbb{R}^d$ is **multivariate normal** with mean $\boldsymbol{\mu}$ and covariance Σ (positive definite), written $\mathbf{X} \sim N_d(\boldsymbol{\mu}, \Sigma)$, if its pdf is

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right).$$

Important properties:

- Any linear combination $\mathbf{a}^\top \mathbf{X}$ is univariate normal.
- Any sub-vector is multivariate normal.
- Conditional distributions are (again) multivariate normal with explicit formulas for the mean and covariance.

Example 7.2 (Bivariate normal regression view). Consider $(X, Y)^\top \sim N_2(\boldsymbol{\mu}, \Sigma)$ where

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}.$$

Then the conditional distribution of Y given $X = x$ is normal with

$$E[Y | X = x] = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X),$$

$$\text{Var}(Y | X = x) = \sigma_Y^2(1 - \rho^2).$$

This is a probabilistic origin of simple linear regression: the conditional mean is a linear function of x with slope $\rho\sigma_Y/\sigma_X$.

8 Conditional Expectation and Laws of Total Expectation/Variance

8.1 Conditional Expectation

Definition 8.1 (Conditional expectation (discrete case)). *For discrete random variables X, Y , the conditional expectation of X given Y is*

$$E[X | Y = y] = \sum_x x P(X = x | Y = y).$$

In the continuous case, replace the sum with an integral involving conditional densities. More abstractly, $E[X | Y]$ is itself a random variable (a function of Y) that averages X over the uncertainty remaining after knowing Y .

Example 8.1 (Tower property intuition). *Let X be daily P&L and Y be a market regime indicator (e.g. bull/bear). Then $E[X | Y]$ is the expected P&L in each regime. If you then average these regime-specific expectations weighted by the probabilities of each regime, you get the unconditional expected P&L:*

$$E[X] = E[E[X | Y]].$$

8.2 Law of Total Expectation

Theorem 8.1 (Law of total expectation). *For random variables X, Y ,*

$$E[X] = E[E[X | Y]].$$

This is sometimes called the “tower property” because applying conditional expectations in nested fashion works like climbing up/down a tower of sigma-algebras.

8.3 Law of Total Variance

Theorem 8.2 (Law of total variance). *For random variables X, Y ,*

$$\text{Var}(X) = E[\text{Var}(X | Y)] + \text{Var}(E[X | Y]).$$

Interpretation: total variability of X is the sum of:

- expected remaining variance of X after conditioning on Y ;
- variance of the conditional means themselves (how different the regimes are).

Example 8.2 (Risk decomposition). *Let X be daily P&L and Y be a volatility regime (low, medium, high). Then*

$$\text{Var}(X) = E[\text{Var}(X | Y)] + \text{Var}(E[X | Y]).$$

The first term represents within-regime risk (how variable P&L is inside each volatility regime), while the second term captures between-regime risk (how different the average P&L is across regimes). This decomposition is useful for understanding where most of your risk comes from.

9 Sampling Distributions: χ^2 , t , and F

9.1 Chi-Square Distribution

Definition 9.1 (Chi-square distribution). If $Z_1, \dots, Z_k$ are independent $N(0, 1)$ random variables, then

$$Q = \sum_{i=1}^k Z_i^2$$

has a **chi-square** distribution with k degrees of freedom, written $Q \sim \chi_k^2$.

Chi-square distributions arise naturally as sums of squared standardized residuals and as sampling distributions of variance estimators under normality.

9.2 Student's t Distribution

Definition 9.2 (Student's t). If $Z \sim N(0, 1)$ and $Q \sim \chi_\nu^2$ are independent, then

$$T = \frac{Z}{\sqrt{Q/\nu}}$$

has a **Student's t** distribution with ν degrees of freedom.

In practice, T appears when we replace the population standard deviation by the sample standard deviation in a normal model.

Example 9.1 (t -statistic for mean). Let $X_1, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$. Then

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a t -distribution with $n - 1$ degrees of freedom, where S^2 is the sample variance. This is the basis of t -tests and t -based confidence intervals for the mean when σ^2 is unknown.

9.3 F Distribution

Definition 9.3 (F distribution). If $Q_1 \sim \chi_{d_1}^2$ and $Q_2 \sim \chi_{d_2}^2$ are independent, then

$$F = \frac{Q_1/d_1}{Q_2/d_2}$$

has an **F distribution** with (d_1, d_2) degrees of freedom.

F distributions appear as ratios of variance estimates and are central to analysis of variance (ANOVA) and multiple regression F -tests.

10 Point Estimation: Properties and Constructions

10.1 Desirable Properties of Estimators

Let θ be a parameter and $\hat{\theta}_n$ an estimator based on a sample of size n .

- **Unbiasedness:** $E[\hat{\theta}_n] = \theta$.
- **Consistency:** $\hat{\theta}_n \xrightarrow{P} \theta$ as $n \rightarrow \infty$.
- **Mean squared error (MSE):** $E[(\hat{\theta}_n - \theta)^2]$.

Unbiasedness is about the average location of the estimator; consistency is about its long-run behavior; MSE balances bias and variance.

Example 10.1 (Biased vs. unbiased variance). For i.i.d. $X_1, \dots, X_n$ with mean μ and variance σ^2 , define

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2, \quad S_n^{2'} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Then $E[S_n^2] = \sigma^2$ (unbiased), but $E[S_n^{2'}] = \frac{n-1}{n} \sigma^2$ (biased). However, both are consistent for σ^2 since the bias of $S_n^{2'}$ shrinks as n grows.

10.2 Method of Moments

The **method of moments** matches sample moments to theoretical moments.

Example 10.2 (Poisson mean by method of moments). Let $X_1, \dots, X_n$ be i.i.d. $\text{Poisson}(\lambda)$. The theoretical mean is $E[X_i] = \lambda$. The sample mean is $\bar{X}$. Method of moments sets

$$\bar{X} = \lambda \quad \Rightarrow \quad \hat{\lambda}_{MM} = \bar{X}.$$

This is intuitive: the average count is a natural estimate of the underlying rate.

10.3 Maximum Likelihood Estimation (MLE)

Let $X_1, \dots, X_n$ be i.i.d. with density/pmf $f(x; \theta)$. The **likelihood** is

$$L(\theta) = \prod_{i=1}^n f(X_i; \theta),$$

and the log-likelihood is $\ell(\theta) = \log L(\theta)$. An MLE $\hat{\theta}_{MLE}$ maximizes $L(\theta)$ (or $\ell(\theta)$).

Example 10.3 (Normal mean and variance). Suppose $X_1, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$ with both μ, σ^2 unknown. The log-likelihood is

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2.$$

Differentiating with respect to μ and setting to zero gives

$$\hat{\mu}_{MLE} = \bar{X}.$$

Differentiating with respect to σ^2 , solving gives

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

So the MLE for μ is the sample mean, and the MLE for σ^2 is the sample variance with divisor n (the biased version but consistent).

10.4 Asymptotic Normality of MLEs

Under regularity conditions, if $\hat{\theta}_{\text{MLE}}$ is the MLE for a scalar parameter θ_0 , then

$$\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta_0) \Rightarrow N\left(0, \frac{1}{I(\theta_0)}\right),$$

where $I(\theta_0)$ is the Fisher information, often defined as

$$I(\theta_0) = E\left[-\frac{\partial^2}{\partial \theta^2} \ell(\theta_0)\right].$$

This yields approximate confidence intervals for θ_0 based on the curvature of the log-likelihood around its maximum.

11 Sufficiency and Exponential Families

11.1 Sufficient Statistics

Definition 11.1 (Sufficient statistic). A statistic $T(X_1, \dots, X_n)$ is **sufficient** for parameter θ if the conditional distribution of the data given T does not depend on θ .

Intuitively, once you know T , the remaining information in the sample about θ is exhausted.

11.2 Factorization Theorem

Theorem 11.1 (Neyman–Fisher factorization). A statistic $T(\mathbf{X})$ is sufficient for θ if and only if the joint density/pmf can be written as

$$f(\mathbf{x}; \theta) = g_\theta(T(\mathbf{x})) h(\mathbf{x}),$$

for some non-negative functions g_θ and h .

Example 11.1 (Normal family). Let $X_1, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$ with σ^2 known. The joint density is

$$f(\mathbf{x}; \mu) \propto \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right).$$

Expanding the square,

$$\sum (x_i - \mu)^2 = \sum x_i^2 - 2\mu \sum x_i + n\mu^2.$$

Thus

$$f(\mathbf{x}; \mu) = h(\mathbf{x}) \cdot g_\mu \left(\sum x_i \right),$$

where $h(\mathbf{x})$ does not depend on μ and g_μ depends on $\mathbf{x}$ only through $\sum x_i$ (equivalently $\bar{X}$). So $\sum X_i$ (or $\bar{X}$) is a sufficient statistic for μ .

11.3 Exponential Families

A family of densities has an **exponential family** form if it can be written as

$$f(x; \theta) = h(x) \exp \left(\eta(\theta)^\top T(x) - A(\theta) \right).$$

Many common distributions (Normal, Poisson, Exponential, Binomial) are exponential families. In such families, $T(x)$ is typically a sufficient statistic, and properties like conjugate priors in Bayesian analysis become simpler.

12 Hypothesis Testing: Neyman–Pearson and Likelihood Ratio Tests

12.1 Neyman–Pearson Lemma (Simple vs. Simple)

Consider testing

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta = \theta_1,$$

where both are simple hypotheses. Let $f(x; \theta_0)$ and $f(x; \theta_1)$ be the corresponding densities.

The Neyman–Pearson lemma states that the most powerful test (for a given level α) rejects H_0 for large values of the likelihood ratio

$$\Lambda(x) = \frac{f(x; \theta_0)}{f(x; \theta_1)}.$$

Equivalently, we can reject for small $\Lambda(x)$; it's common to work with $-2 \log \Lambda$.

12.2 Likelihood Ratio Tests (LRTs)

For general tests $H_0 : \theta \in \Theta_0$ vs. $H_1 : \theta \in \Theta_1$, define the likelihood ratio statistic

$$\Lambda(x) = \frac{\sup_{\theta \in \Theta_0} L(\theta)}{\sup_{\theta \in \Theta} L(\theta)}.$$

We reject H_0 for small values of $\Lambda(x)$.

Under regularity conditions, and when H_0 is true, Wilks' theorem says that

$$-2 \log \Lambda(X) \stackrel{\text{approx}}{\sim} \chi_k^2,$$

where k is the difference in dimensionality between Θ and Θ_0 .

Example 12.1 (Testing a normal mean, variance unknown). *Let $X_1, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$ with σ^2 unknown. We test $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$. The full MLEs $(\hat{\mu}, \hat{\sigma}^2)$ maximize the likelihood over both parameters; under H_0 , we fix $\mu = \mu_0$ and maximize over σ^2 only. Computing $-2 \log \Lambda$ leads (after algebra) to a function of the t -statistic*

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}.$$

Thus the classical t -test can be seen as a likelihood ratio test in this normal model.

13 Bayesian Inference Basics

13.1 Bayes' Rule for Parameters

In Bayesian inference, we treat the parameter θ as a random variable with prior density $\pi(\theta)$. Given data x , the posterior density is

$$\pi(\theta | x) = \frac{f(x | \theta) \pi(\theta)}{\int f(x | \theta') \pi(\theta') d\theta'}.$$

The numerator is the unnormalized posterior, and the denominator is the marginal likelihood $f(x)$.

13.2 Conjugate Priors

A **conjugate prior** is one for which the posterior distribution is in the same family as the prior.

Example 13.1 (Binomial-Beta model). *Let $X | p \sim \text{Binomial}(n, p)$ and prior $p \sim \text{Beta}(\alpha, \beta)$. The likelihood is*

$$f(x | p) \propto p^x (1 - p)^{n-x},$$

and the prior density is

$$\pi(p) \propto p^{\alpha-1} (1 - p)^{\beta-1}.$$

Multiplying,

$$\pi(p | x) \propto p^{x+\alpha-1} (1 - p)^{n-x+\beta-1},$$

which is a $\text{Beta}(\alpha + x, \beta + n - x)$ density. So the Beta prior is conjugate to the Binomial likelihood.

13.3 Posterior Mean and Credible Intervals

A common Bayesian point estimate is the **posterior mean** $E[\theta | x]$. Intervals (a, b) with

$$P(a < \theta < b | x) = 0.95$$

are called 95% **credible intervals**.

Example 13.2 (Posterior mean in Binomial-Beta). *In the Beta-Binomial example, if the prior is $\text{Beta}(\alpha, \beta)$ and we observe x successes out of n , the posterior is $\text{Beta}(\alpha + x, \beta + n - x)$. The posterior mean is*

$$E[p | x] = \frac{\alpha + x}{\alpha + \beta + n}.$$

This can be viewed as a weighted average of the prior mean and the sample proportion x/n , with weights depending on the prior strength $(\alpha + \beta)$ vs. data size n .

14 Multiple Linear Regression in Matrix Form

14.1 Model and Least Squares Estimator

The multiple linear regression model is

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where:

- $\mathbf{y}$ is an $n \times 1$ response vector.
- X is an $n \times p$ design matrix (first column often ones for intercept).
- $\boldsymbol{\beta}$ is a $p \times 1$ vector of coefficients.
- $\boldsymbol{\varepsilon}$ is an $n \times 1$ error vector.

The least squares estimator solves

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \|\mathbf{y} - X\boldsymbol{\beta}\|^2.$$

Assuming $X^\top X$ is invertible,

$$\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}.$$

14.2 Gauss–Markov Theorem

Assume the errors have $E[\boldsymbol{\varepsilon}] = 0$ and $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 I_n$ (homoskedastic and uncorrelated). Then $\hat{\boldsymbol{\beta}}$ is unbiased and has covariance

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (X^\top X)^{-1}.$$

The Gauss–Markov theorem states that among all linear unbiased estimators of $C\boldsymbol{\beta}$ (for any fixed matrix C), $C\hat{\boldsymbol{\beta}}$ has the smallest variance. So ordinary least squares (OLS) is the best linear unbiased estimator (BLUE).

Example 14.1 (Predicting returns from factors). *Let $\mathbf{y}$ be the vector of daily excess returns of a stock, and X the matrix of factor returns (e.g., market, size, value factors). The model*

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

estimates the factor loadings $\boldsymbol{\beta}$. The OLS estimate $\hat{\boldsymbol{\beta}}$ gives the best linear unbiased estimate of these loadings under the usual regression assumptions, and the covariance matrix $\hat{\sigma}^2 (X^\top X)^{-1}$ quantifies uncertainty in the loadings.

14.3 Inference and F-tests

Under the stronger assumption $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 I_n)$, we can derive exact sampling distributions:

- $\hat{\boldsymbol{\beta}}$ is multivariate normal.
- $S^2 = \frac{1}{n-p} \|\mathbf{y} - X\hat{\boldsymbol{\beta}}\|^2$ is an unbiased estimator of σ^2 .

Individual coefficients can be tested via t-statistics; sets of coefficients (e.g., testing whether a group of predictors jointly has zero effect) can be tested via F-statistics constructed from sums of squared residuals under full and reduced models.

15 Logistic Regression Overview

15.1 Binary Response Model

When the response Y is binary (0/1), a linear model for $E[Y \mid \mathbf{X}]$ is often inappropriate because it can produce values outside $[0, 1]$. Logistic regression models the probability

$$\pi(\mathbf{x}) = P(Y = 1 \mid \mathbf{X} = \mathbf{x})$$

via the **logit** link:

$$\log \frac{\pi(\mathbf{x})}{1 - \pi(\mathbf{x})} = \boldsymbol{\beta}^\top \mathbf{x}.$$

15.2 Interpretation

Solving for $\pi(\mathbf{x})$ gives

$$\pi(\mathbf{x}) = \frac{\exp(\boldsymbol{\beta}^\top \mathbf{x})}{1 + \exp(\boldsymbol{\beta}^\top \mathbf{x})}.$$

Each coefficient β_j represents the change in the log-odds per unit increase in x_j holding other variables fixed. Exponentiating β_j gives the odds ratio.

Example 15.1 (Default probability). *Let Y indicate whether a borrower defaults (1) or not (0), and let X_1 be the debt-to-income ratio, X_2 a credit score. A logistic regression model*

$$\log \frac{\pi}{1 - \pi} = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

relates log-odds of default to these predictors. If $\hat{\beta}_1 = 0.05$, then increasing X_1 by 1 unit multiplies the odds of default by $\exp(0.05) \approx 1.051$, or about a 5.1% increase in the odds, holding credit score fixed.

16 Resampling and the Bootstrap

16.1 Motivation

Analytical formulas for the sampling distribution of an estimator can be complex or unavailable, especially for non-standard statistics or in small samples. The **bootstrap** approximates the sampling distribution by resampling from the data.

16.2 Nonparametric Bootstrap Algorithm

Given data $X_1, \dots, X_n$ and an estimator $\hat{\theta} = t(X_1, \dots, X_n)$:

1. For $b = 1, \dots, B$ (e.g., $B = 1000$):
 - (a) Draw a bootstrap sample $X_1^{*(b)}, \dots, X_n^{*(b)}$ by sampling with replacement from $\{X_1, \dots, X_n\}$.
 - (b) Compute the bootstrap replicate $\hat{\theta}^{*(b)} = t(X_1^{*(b)}, \dots, X_n^{*(b)})$.
2. Use the empirical distribution of $\{\hat{\theta}^{*(b)}\}_{b=1}^B$ to approximate the sampling distribution of $\hat{\theta}$.

16.3 Bootstrap Confidence Intervals

A simple percentile bootstrap interval for θ at level $(1 - \alpha)$ is

$$\left(\hat{\theta}^{*(\alpha/2)}, \hat{\theta}^{*(1-\alpha/2)} \right),$$

where $\hat{\theta}^{*(q)}$ is the empirical q -quantile of the bootstrap replicates.

Example 16.1 (Bootstrap CI for median). *Suppose $X_1, \dots, X_n$ are asset returns, and we are interested in the true median return θ . The sample median $\hat{\theta}$ has a complicated sampling distribution, especially for small n . The bootstrap approach:*

1. *Resample with replacement from the observed returns to create many bootstrap samples.*
2. *Compute the median for each sample.*
3. *Take the empirical 2.5% and 97.5% quantiles of these medians as a 95% confidence interval.*

This gives a data-driven uncertainty quantification without relying on an explicit parametric model for returns.

17 Next Steps and Further Reading

This intermediate course has introduced:

- More formal probability foundations and modes of convergence.
- Asymptotic tools (CLT, delta method, mgfs/characteristic functions).
- Multivariate distributions and conditional expectation.
- Sampling distributions, point estimation (MLE, method of moments), sufficiency.
- Classical hypothesis testing, Bayesian basics, and regression at a matrix level.
- A first look at logistic regression and resampling methods like the bootstrap.

Natural next directions include:

- Advanced likelihood theory and information inequalities.
- Nonparametric estimation and smoothing methods.
- Time series models (ARMA, GARCH) and stochastic processes.
- Causal inference and experimental design.

Courses labeled “mathematical statistics” or “fundamentals of statistics” at universities often cover comparable material and provide problem sets and exam-style practice to consolidate these ideas.²

²See, for example, MIT 18.650, various “Intermediate Statistics” and “Mathematical Statistics” syllabi.[\[web:32\]](#)[\[web:36\]](#)[\[web:38\]](#)[\[web:40\]](#)[\[web:41\]](#)